

Supplementary Material for Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures

Simone Croci, Athanasios Charisoudis, Kit Yung Lam and Aljosa Smolic
Lucerne University of Applied Sciences and Arts, Rotkreuz, Switzerland
Email: simone.croci@hslu.ch

This document contains the supplemental material of the paper *Evaluations of Dynamic Gaussian Splats versus Point Clouds for Sparse Captures* (S. Croci et al) [1].

I. STIMULI FROM THE GAUSSIAN SPLAT VS POINT CLOUD EXPERIMENT

Figure 1 shows a sample frame from the stimuli of the subjective experiment with point cloud and Gaussian splat reconstructions on the left and right, respectively. In the captions of this figure, there are the names of the subjects that were reconstructed. In the experiment, we also used the same stimuli with the reversed order of the reconstructions.

II. PER-DYNAMIC-RECONSTRUCTION GAUSSIAN SPLAT ANALYSIS

This section presents the GS analysis of all reconstructions in Figure 2 and then the per-sequence analysis in the Figures 3, 4, 5, 6, 7, 8. As can be observed, the analysis of the reconstructions do not vary so much.

III. ANALYSIS OF THE STIMULI WATCHING TIME

This section presents the analysis of the stimuli watching times collected during the subjective experiment. In particular, Figure 9 shows the histogram of the watching times. In the histogram, there is a peak at 10 s, i.e., the duration of the stimuli.

IV. STATIC GAUSSIAN SPLAT RECONSTRUCTIONS ANALYSIS

This section presents the GS analysis of the static GS reconstructions presented in the paper *Gaussian Splatting vs. Classical Photogrammetry: A Comparison for Virtual Backdrops* (P. Haslbauer et al.) [2]. These reconstructions were generated with a scene GS optimization method called Postshot [3]. Figure 10 shows the GS analysis of all static reconstructions. It can be observed, that there is a clear difference between the dynamic GS reconstructions of our paper and the static ones of [2].

REFERENCES

- [1] A. Charisoudis, S. Croci, K. Y. Lam, P. Frossard, and A. Smolic, "A fast volumetric capture and reconstruction pipeline for dynamic point clouds and gaussian splats," in *Proceedings of the 22nd ACM SIGGRAPH European Conference on Visual Media Production*, ser. CVMP '25. New York, NY, USA: Association for Computing Machinery, 2025. [Online]. Available: <https://doi.org/10.1145/3756863.3769713>
- [2] P. Haslbauer, M. Pullen, C. Reichherzer, and A. Smolic, "Gaussian splatting vs. classical photogrammetry: A comparison for virtual backdrops," in *2025 17th International Conference on Quality of Multimedia Experience (QoMEX)*. IEEE, 2025, pp. 1–4.
- [3] (2025) Postshot. [Online]. Available: <https://www.jawset.com/>

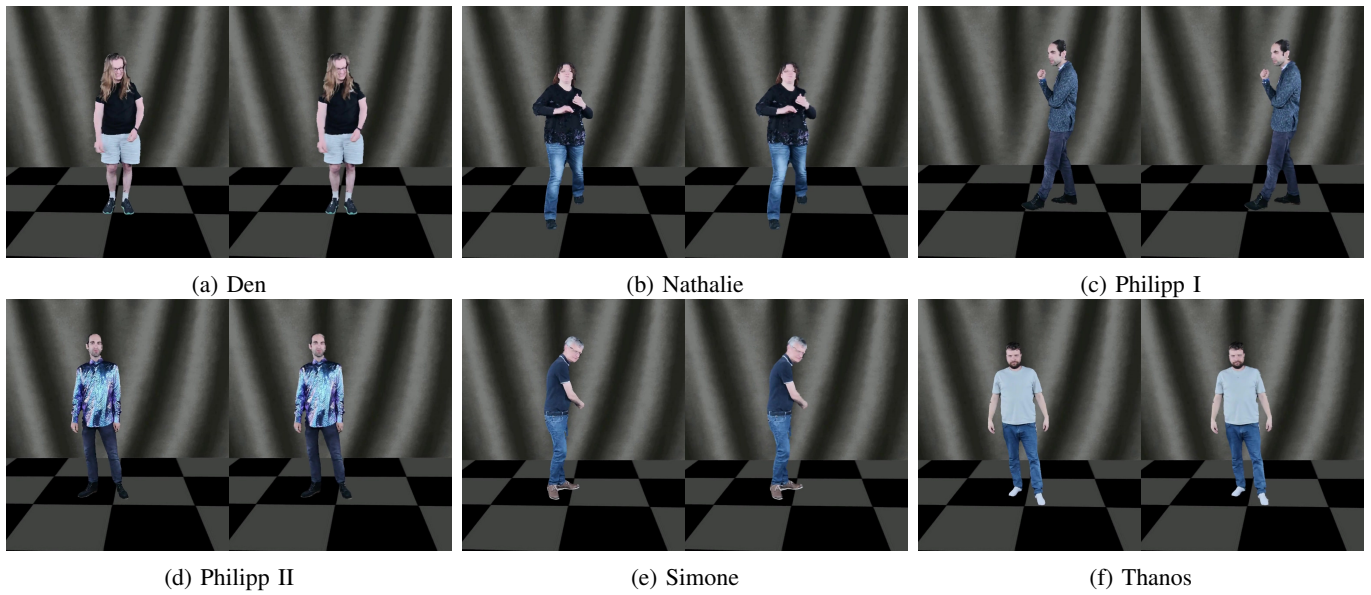


Fig. 1: Stimuli with point cloud and Gaussian splat reconstructions on the left and right, respectively. In the experiment, we also use the same reconstructions but with reversed order.

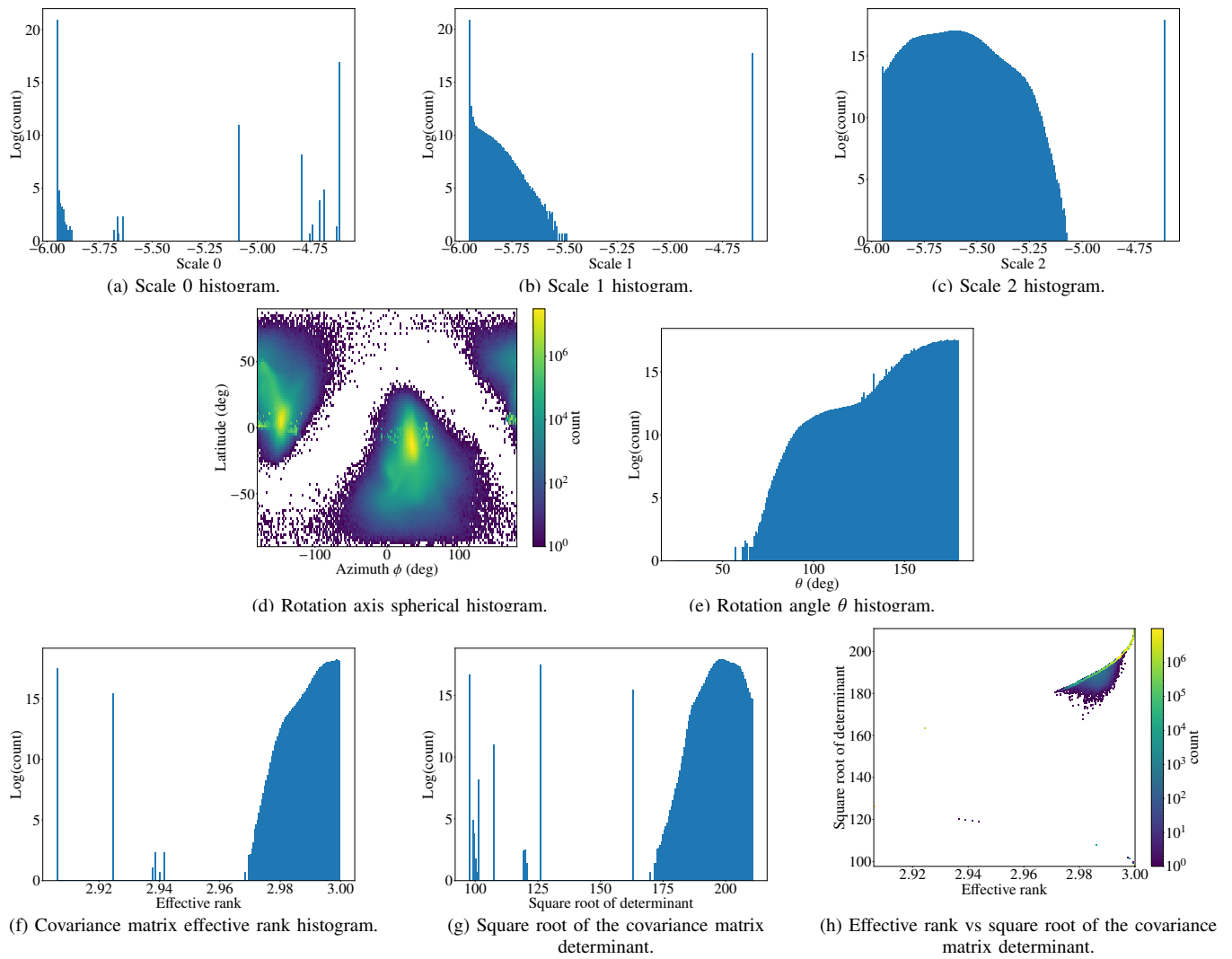


Fig. 2: Gaussian splats analysis of all dynamic reconstructions.

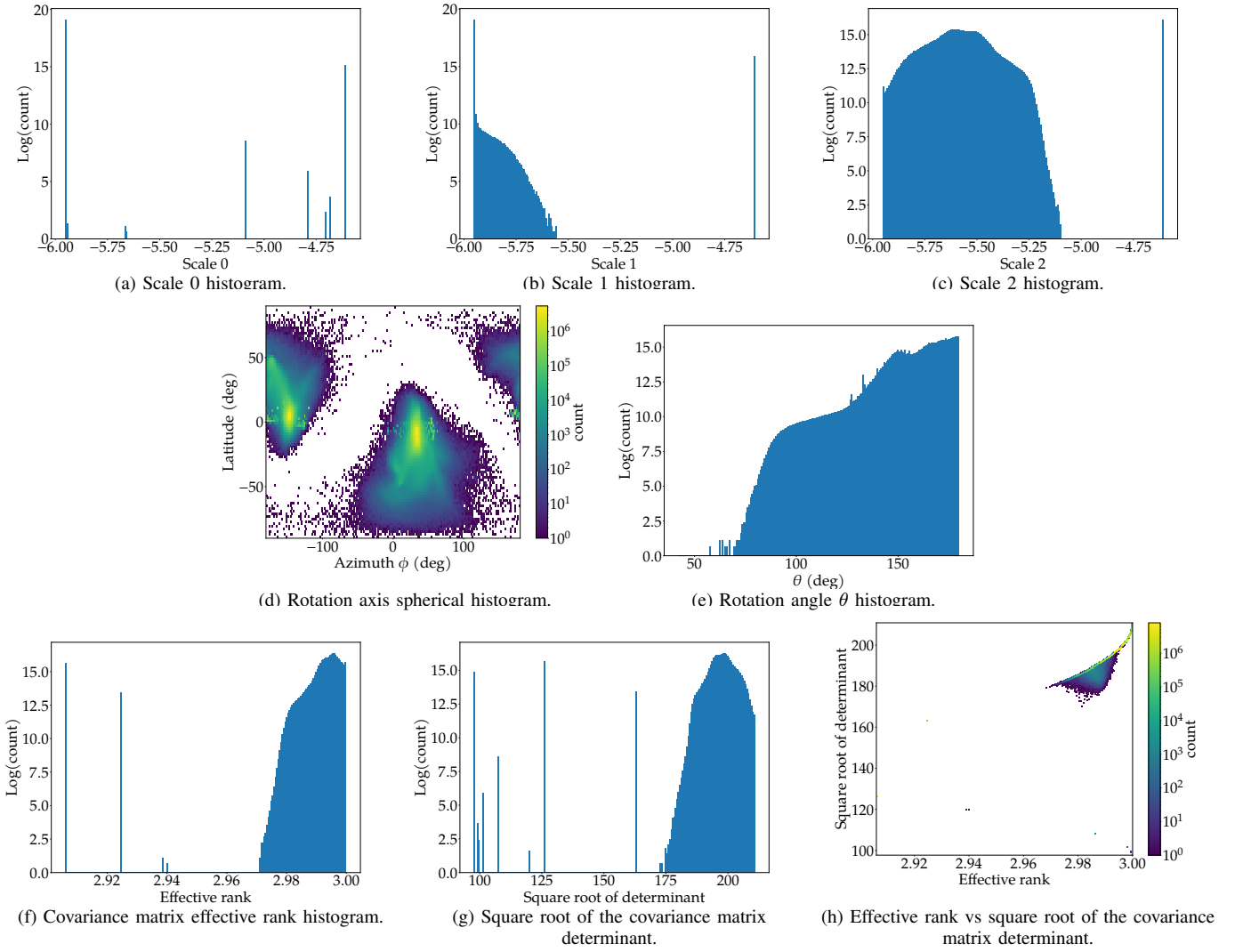


Fig. 3: Gaussian splats analysis of Den's dynamic reconstruction.

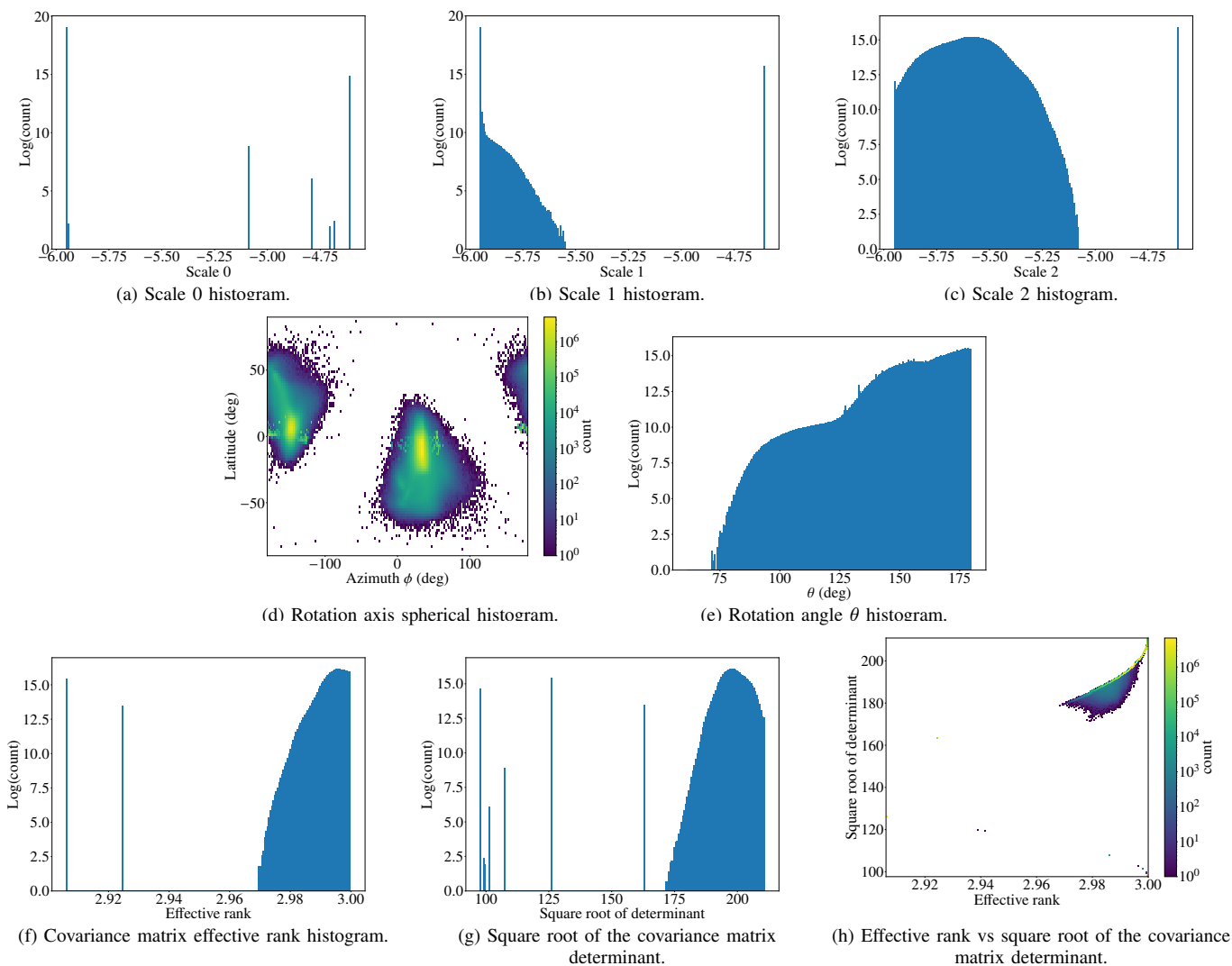


Fig. 4: Analysis of the Gaussian splats from Nathalie's dynamic reconstruction.

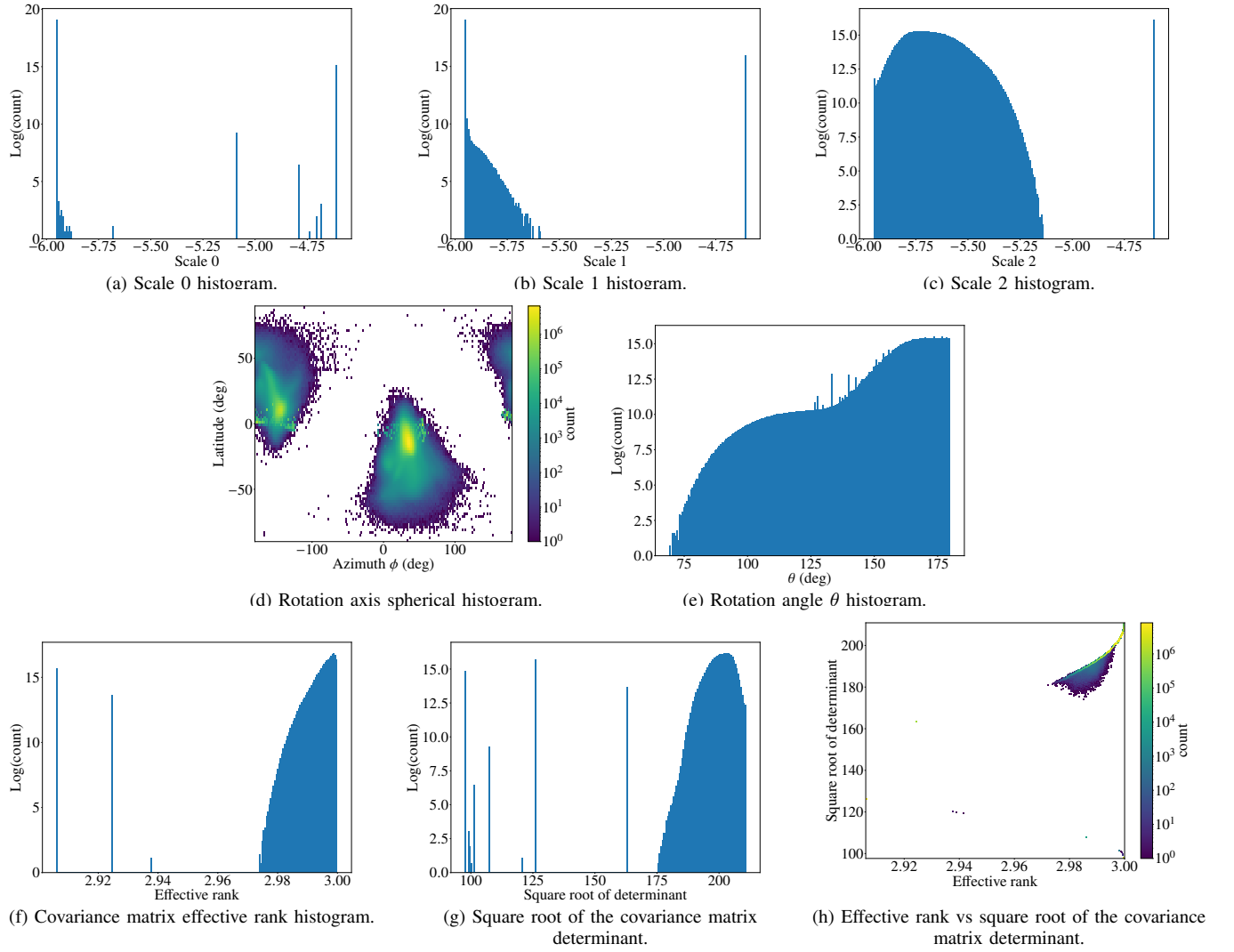
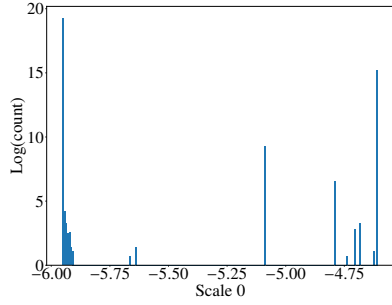
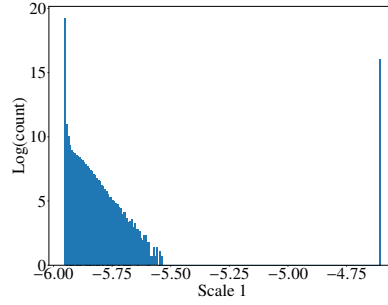


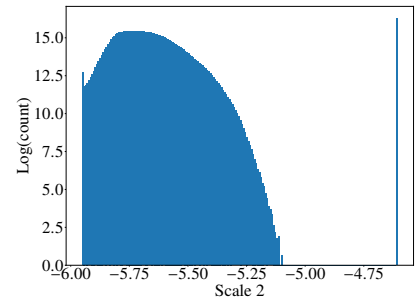
Fig. 5: Gaussian splats analysis of Philipp I's dynamic reconstruction.



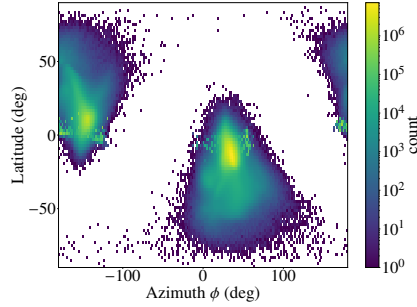
(a) Scale 0 histogram.



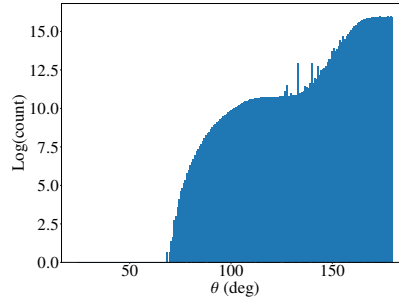
(b) Scale 1 histogram.



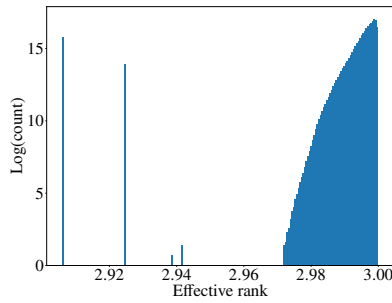
(c) Scale 2 histogram.



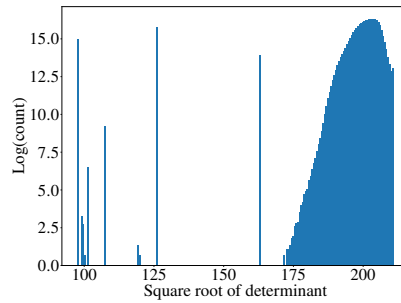
(d) Rotation axis spherical histogram.



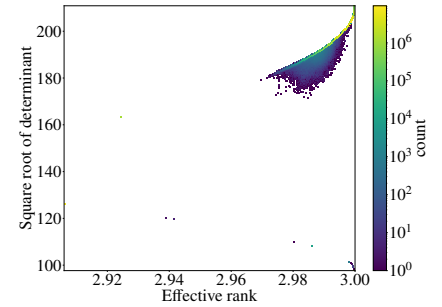
(e) Rotation angle θ histogram.



(f) Covariance matrix effective rank histogram.



(g) Square root of the covariance matrix determinant.



(h) Effective rank vs square root of the covariance matrix determinant.

Fig. 6: Gaussian splats analysis of Philipp II's dynamic reconstruction.

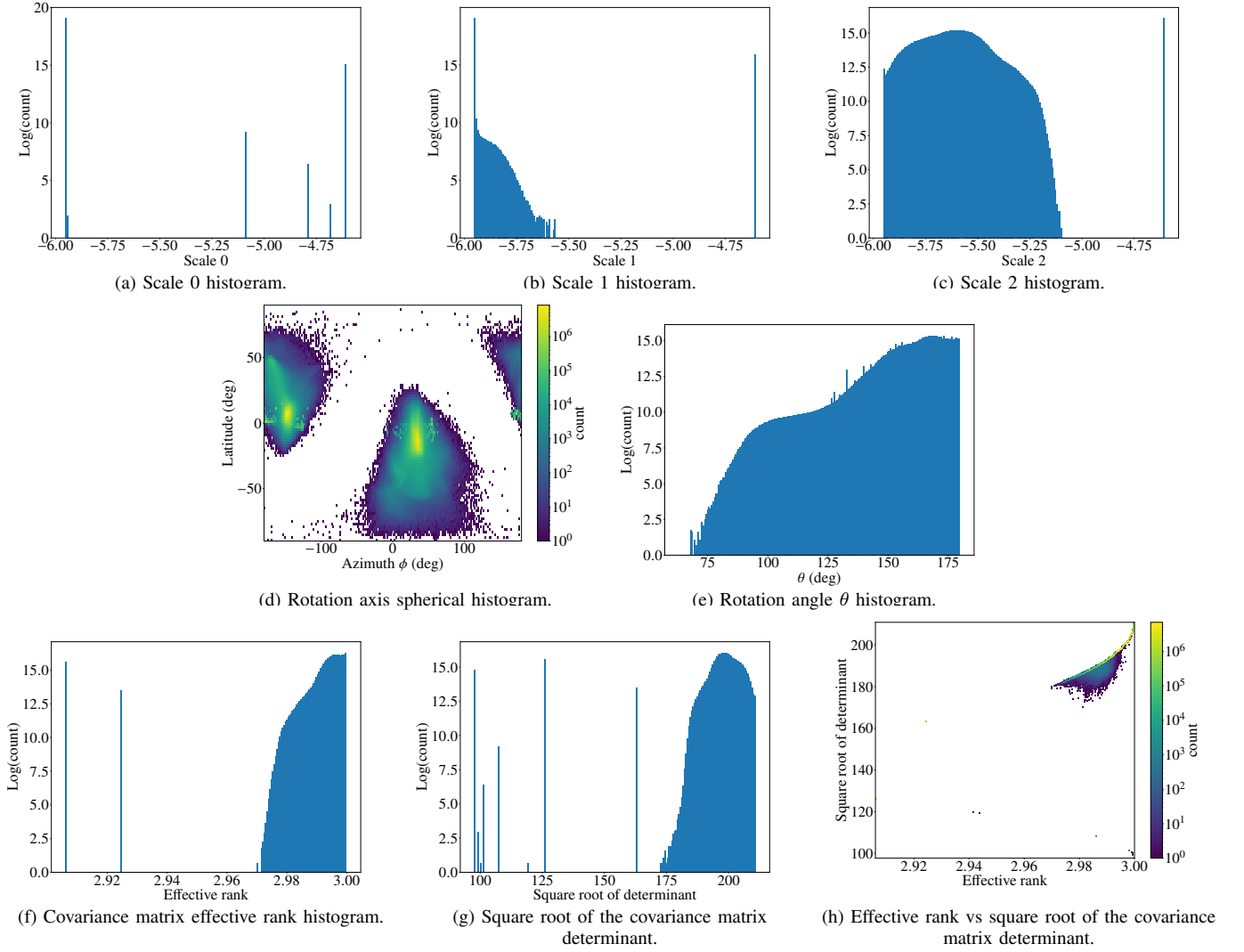


Fig. 7: Gaussian splats analysis of Simone's dynamic reconstruction.

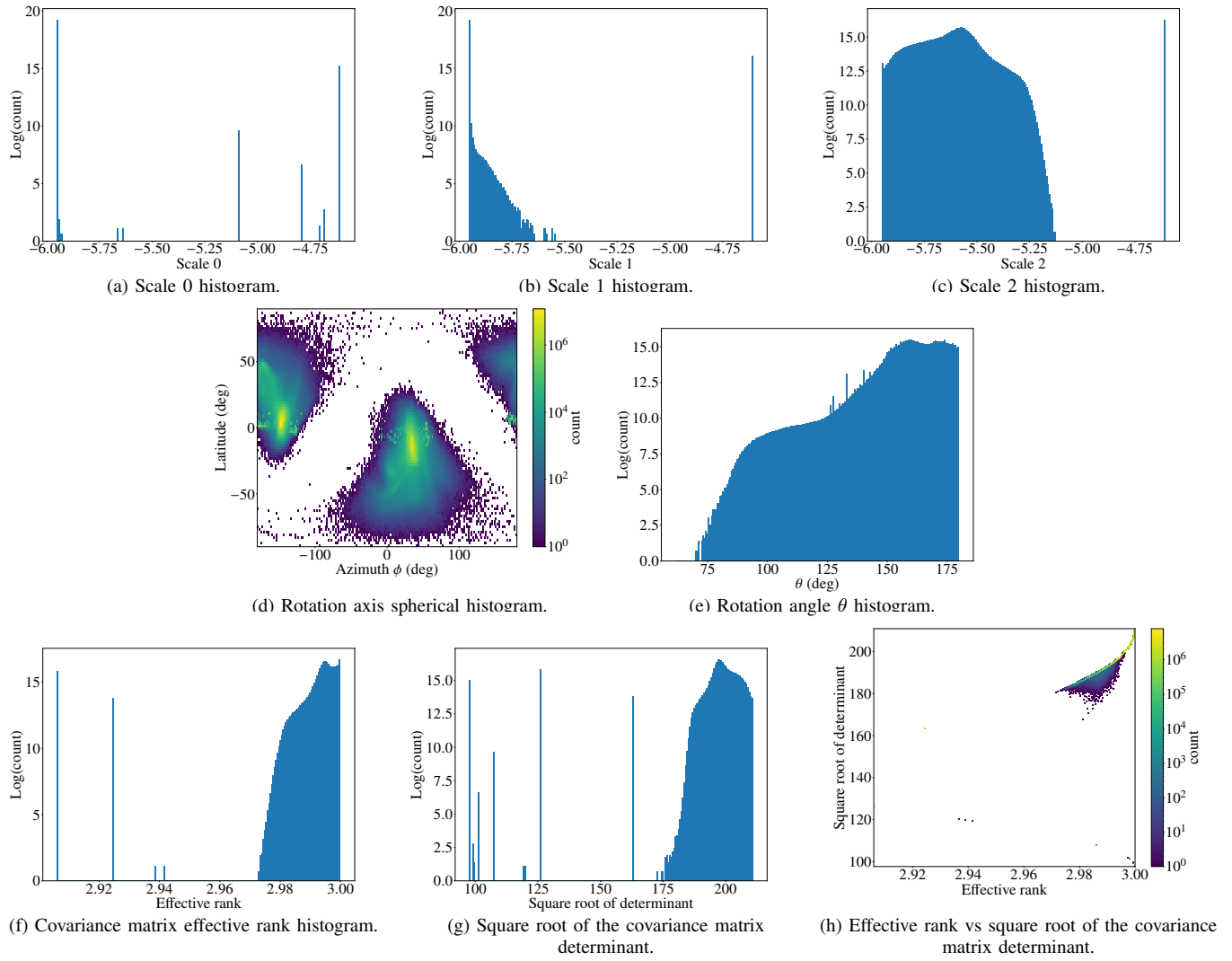


Fig. 8: Gaussian splats analysis of Thanos's dynamic reconstruction.

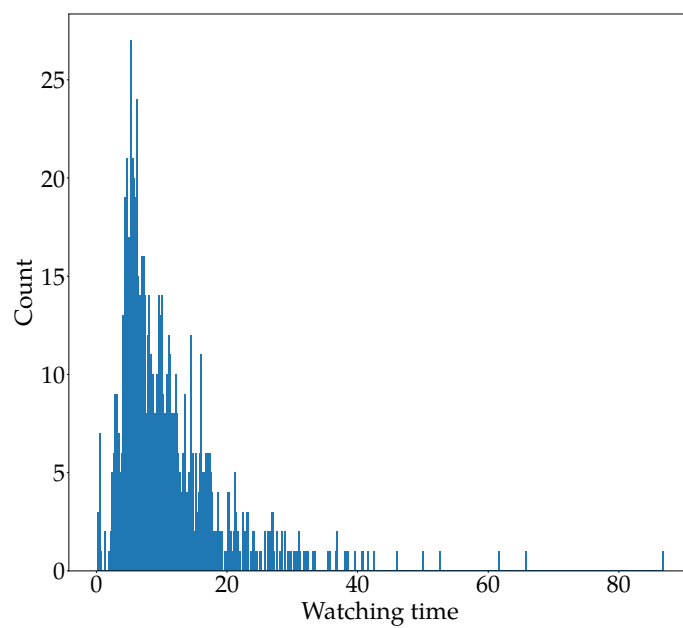


Fig. 9: Stimuli watching time histogram.

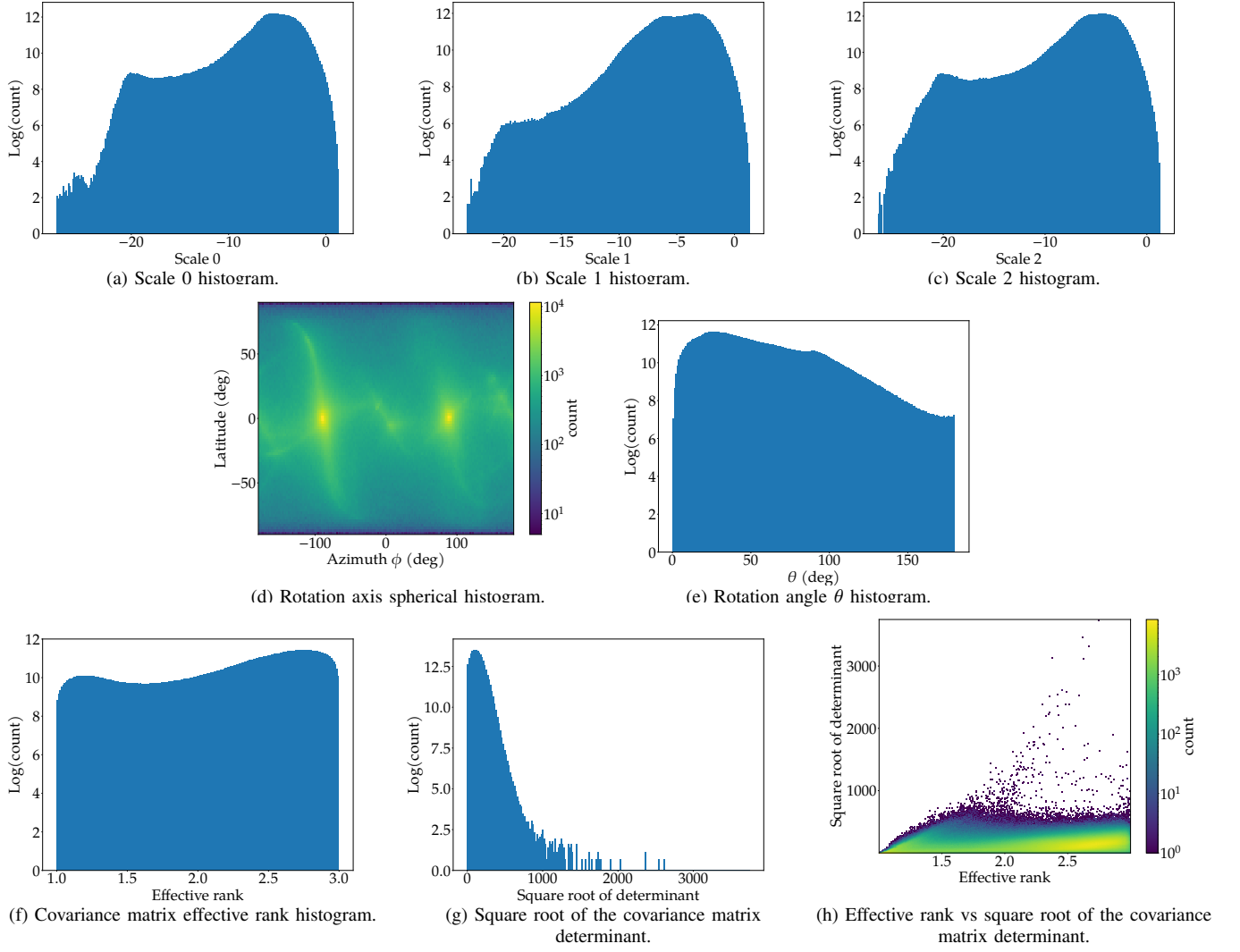


Fig. 10: Gaussian splats analysis from all static reconstructions.